Technical Proposal

Cost-Aware Design of Seawater-Resistant Copper Alloy Powders for Blown-Powder Directed Energy Deposition

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1. Technical Approach

1.1 Problem and objectives

Much of the Navy's seawater service runs on two kinds of copper alloy. Copper-nickel alloys, 90-10 (UNS C70600) and 70-30 (UNS C71500), carry seawater in piping, condensers, and heat exchangers, and nickel-aluminum bronze (NAB, UNS C95800) is cast into propellers, valves, and pump parts [1, 2, 3]. Both were developed for casting or wrought processing. Directed energy deposition (DED), in which a laser melts a stream of powder blown into a moving melt pool, can build large parts and restore worn or damaged ones without a foundry [4, 5]. DED, however, cools each layer from the melt within seconds and reheats it many times as later layers pass, so it produces microstructures these alloys were never designed for. In NAB, fast cooling can retain a martensitic phase that corrodes selectively in seawater [6, 7], and the Navy's specification for NAB deposited by wire-arc DED requires the same 6-hour anneal as castings [8]. That step is costly for a large part and impractical for many repairs made in place. The Navy's specifications for printed copper alloys all use existing compositions [8, 9, 10], and the design of compositions for laser deposition has only begun [11].

Designing one runs through a chain of steps: a thermodynamic calculation, a powder, deposited test specimens, microscopy, and property tests. The costs of the steps differ greatly. A thermodynamic calculation takes minutes, a powder lot a day, a DED session days including turnaround, and a corrosion immersion weeks; the protective film that gives copper-nickel its resistance takes weeks to months to mature [12, 13]. Experimental campaigns often send each candidate through the whole chain, spending much of the budget on full tests of candidates that a cheap early measurement could have ruled out. Computational campaigns often work as a funnel: many simulations, then a few candidates passed on to experiments. If the simulation misses a physical effect, the funnel discards good candidates before any experiment can catch the error.

The powder is a second obstacle. Single-stream blown-powder DED, the form in use today, feeds a pre-alloyed powder from one hopper, so each new composition needs its own powder lot, in a size range that flows through the nozzle and that the melt pool catches [14, 15]. The lot must also fill the hopper, about 2 kg on the system this project will use (Section 1.8.2). Industrial gas atomizers make powder in lots of tens of kilograms or more, so new compositions are rarely tried [TODO Sterling: cite a minimum lot size, e.g., a vendor's]. Our ultrasonic atomizer makes a lot of a new composition in a day from about half a kilogram of feedstock, and 2 kg in about six runs. Ultrasonic atomizers have already made powder for blown-powder DED from NAB propeller grinding chips [16, 17] and from copper cable scrap [18], but only in existing compositions. That turns the powder from a purchase into steps the search can choose: a small lot for cheap tests of many compositions, and a full lot only for those worth depositing. It also lets the search design the powder along with the alloy.

This project asks a basic question. When the steps of a materials workflow differ in cost by orders of magnitude, and results arrive at different times, how should one choose the next measurement and the next candidate? It answers the question on a concrete design problem: a pre-alloyed copper alloy powder for single-stream blown-powder DED. The powder must deposit without cracks or pores, on itself and on the alloys it would repair, and the deposit must resist seawater without heat treatment at least as well as 70-30 copper-nickel. The project has five objectives:

1. Extend a cost-aware search method that learns from partial results to several competing objectives: dense and crack-free deposits, strength, ductility, and resistance to corrosion in seawater.
2. Extend it to many experiments in progress at once, with results that return at different times.
3. Give it physics computed for the process, thermodynamic calculations evaluated along the thermal history of a DED deposit, and measure what that physics is worth.
4. Demonstrate the method on a physical workflow that designs copper alloy powders for blown-powder DED, from powder to deposit, repair coupon, tensile test, and seawater corrosion test, and test whether its stated confidence in a candidate matches what the deposits and tests later show.
5. Measure how well a one-hour electrochemical screen predicts immersion tests that take weeks, so that the long tests run only where they can change a decision.

1.2 Copper alloys in naval seawater service

Navy specifications cover copper-nickel tube for the piping systems of surface ships and submarines [1] and NAB castings for seawater service [2], and NAB is widely used for valve bodies, pump casings, and impellers, with a large market in submarine fleets [3]. Copper-nickel owes its resistance to a thin, adherent film of corrosion products, mainly cuprous oxide with nickel and iron. The film builds up over weeks to months of exposure [12]; in one study the films on 90-10 tubes in natural seawater were still changing after six months [13]. The iron content of the alloy changes the makeup of the film [19]. NAB is a cast alloy of copper with about 9% aluminum and 4 to 5% each of iron and nickel (by weight). Its microstructure of α grains and several intermetallic κ phases resists seawater well, but the α phase in the lamellar $\alpha + \kappa$ eutectoid, and any retained β phase, corrode selectively, above all in crevices [20, 21]. Navy NAB castings therefore get a temper anneal at $1250 \pm 50\,^{\circ}\mathrm{F}$ (about $677\,^{\circ}\mathrm{C}$) for at least 6 hours, propellers excepted [2], which improves their corrosion resistance [3].

Printing changes these microstructures, and printing changes how alloys corrode [22]. How NAB is processed sets its microstructure [23], and NAB has now been printed by laser powder bed fusion [24], wire-arc deposition [25, 26], and laser DED [6, 27]. Laser-deposited NAB contains martensitic β' along with α and κ phases [6, 27] and is much stronger than cast NAB [27]. Its corrosion depends on the process. NAB from laser powder bed fusion formed stronger films than wrought NAB but resisted localized corrosion worse, and annealing reversed both [28]; wire-arc NAB corroded in its residual β phase [29]. Whether annealing helps depends on the starting microstructure. After 480 hours in salt solution, the same $700\,^{\circ}\mathrm{C}$ anneal had lowered the corrosion current of cast NAB and raised that of sintered and hot isostatically pressed NAB, each about threefold, though the authors call the differences small [30]. As-built wire-arc NAB resisted corrosion better than annealed coupons, in a test with unequal immersion times [31]. Laser-deposited NAB on stainless steel was attacked to a depth of 3.4 μm in 21 days in salt solution where iron from the steel had diluted it, and 1.8 μm over a nickel-chromium interlayer [32]. In welded NAB, retained β' in the heat-affected zone corroded selectively in seawater [7], a warning for repairs. Laser cladding of cast NAB has used an NAB consumable [33], and DED repair of cast NAB a low-nickel copper alloy powder [34, 35], later blended in the melt pool with an aluminum alloy powder to vary the aluminum content [36].

In wire-arc 70-30 copper-nickel, the dendrite cores hold more nickel than the regions between them, and those regions corrode first [37]. The corrosion of powder-bed 70-30 has also been studied [38], and its resistance depends on the laser settings [39]. At their best settings, both wire-arc and powder-bed 70-30 have matched or beaten wrought plate in electrochemical tests in salt solution [37, 39]. Laser cladding of Cu-30Ni dates to 2005 [40], and later blown-powder DED of copper-nickel has mixed copper and nickel-alloy powders in the melt pool to make graded layers and thermoelectric alloys [41, 42, 43]. The one corrosion study we found of copper-nickel deposited by blown-powder laser DED is from 2011: Cu-30Ni made by CO_2-laser direct metal deposition on wrought 70-30 plate [44]. A later review reports that the deposit corroded faster than the plate and attributes this to its porosity [45] [TODO Sterling: check the 2011 article before citing a direction. Read literally, its abstract ranks the deposit as the most corrosion resistant and the plate as the least, the reverse of the review. A secondary source found by web search quotes ASTM G31 rates of 0.36 mpy for the deposit and 0.08 mpy for the plate. Neither Edison nor this runner could open the article]. NAVSEA has written specifications for 70-30 copper-nickel made by wire-arc DED and by laser powder bed fusion, and for NAB made by wire-arc DED; the wire-arc ones are alternatives to castings, and all three keep the existing compositions [8, 9, 10]. We found none for either alloy made by blown-powder laser DED. In 2023 a Navy official said, as Defense News reported, that for naval metals "including copper-nickel," "more research is needed" on how printing "affects the metals' fatigue and corrosion properties" [46]. A composition chosen for casting is not necessarily the best one for DED, and each composition and set of deposition settings needs its own tests.

Corrosion tests of copper alloys follow the same cost pattern as the rest of the workflow. Open-circuit potential, linear polarization, and impedance measurements in substitute ocean water [47] take about an hour and give a corrosion rate and a measure of the surface film [48, 49]. Immersion tests with mass loss [50, 51] and cross-sections that measure the depth of selective attack take weeks. Tests in flowing seawater, which set the velocity limits used in piping design, take months. The search can therefore treat corrosion the way it treats cracking and strength: a cheap screen for many candidates and long tests for a few. The cheap screen can mislead, though. An hour-long measurement on a fresh surface sees neither the film that forms over weeks on copper-nickel nor slow selective attack in NAB. On wrought 70-30 copper-nickel in synthetic seawater, the corrosion current fell more than eightfold over the first ten days [52], and welded NAB corroded faster in natural seawater than in simulated seawater [53]. Knowing when the screen can be trusted is part of the research.

Supply and melting add one more objective. Nickel has been on the U.S. list of critical minerals since 2022, and copper joined it in 2025 [54, 55]. Nickel also raises the melting range: 70-30 copper-nickel melts between about 1,171 and 1,238 °C, 90-10 between 1,099 and 1,149 °C, and NAB between 1,043 and 1,060 °C [56]. A replacement for 70-30 copper-nickel that needs less nickel would use less of a critical mineral and would be easier to atomize and deposit. The search will carry nickel content as an objective to minimize where the corrosion trade-offs allow.

1.3 Scientific concepts

The search method treats the manufacturing workflow as a network of linked models, a function network [57]. Each step gets its own statistical model, a Gaussian process, which predicts the outputs of that step together with their uncertainty. The models are linked in the same order as the steps (Fig. 1), so the outputs of one step are inputs to the next and uncertainty carries through the

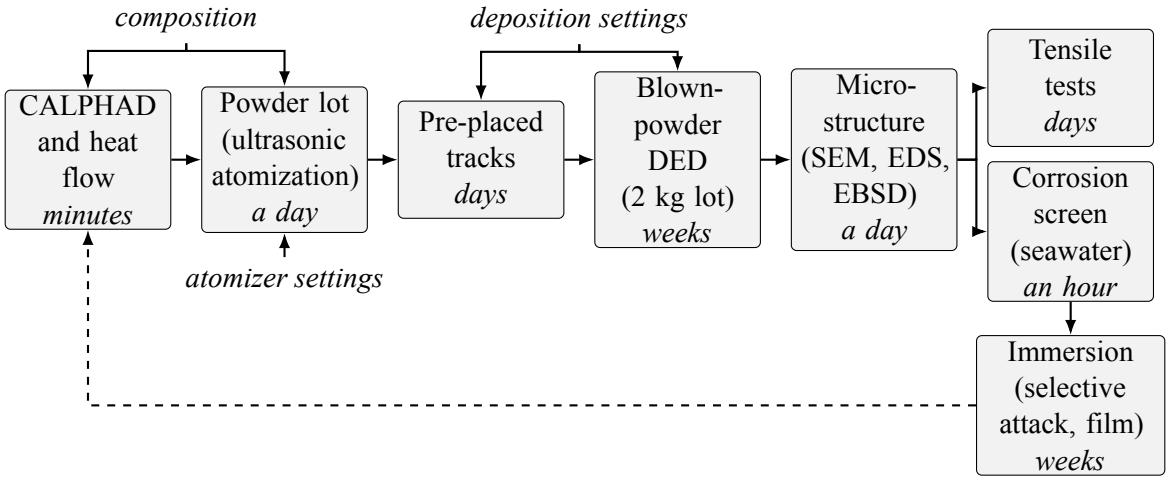


Figure 1. The workflow as linked models, with typical times per result in italics. Pre-placed tracks use grams of a first powder lot. A composition that passes is scaled up to the 2 kg lot that the hopper of the DED system needs, so its blown-powder deposits (tracks, walls, blocks, and repair coupons) take weeks, including the extra atomizer runs and turnaround at the facility. Composition enters at the first two steps, atomizer settings at the second, and deposition settings (laser power, travel speed, and for blown powder the feed rate) at the third and fourth; the heat flow model also takes the deposition settings. After microscopy the network branches to tensile tests and to corrosion tests. Dashed arrow: each result updates the models, which choose the next experiments and steps by expected gain per cost.

whole network. Because each step has its own model, the network learns from partial results, and a candidate can stop at any step. Buathong et al. added a cost-aware rule to this network [58]. For each possible experiment, the rule estimates how much the result is expected to improve the search, divides that by the experiment's cost, and runs the best one.

The physics comes first, in the manner of integrated computational materials engineering (ICME) [59, 60]: calculations decide where experiments go, and experiments correct the calculations. Calculation of phase diagrams (CALPHAD) in Thermo-Calc and pycalphad [61, 62] gives the liquidus and freezing range of each composition and an index of how prone it is to cracking as it solidifies [63]. Such indices have been checked against printed aluminum alloys, steels, and nickel superalloys, but not copper alloys [64]. It also gives how nickel, iron, and aluminum divide between the dendrite cores and the last liquid to freeze, and the phases stable at each temperature, including the β and κ phases of the bronzes. DED adds a thermal history. A moving point-source model of heat flow [65, 66], with temperature-dependent properties of copper-nickel and NAB [67, 68, 69] and calibrated to measured melt-pool sizes, gives for each set of deposition settings the cooling rate through solidification and the peak temperatures that later passes impose on each layer. The CALPHAD calculations are then evaluated at those temperatures, to estimate which phases can form as the deposit solidifies and which can form or dissolve as it is reheated. These outputs enter the statistical models as inputs with learned weights, so the search can discount any of them that proves uninformative. The calculations run continuously on BYU high-performance computing, and each new result is available to the next decision.

1.4 Comparison with other approaches

To our knowledge, neither a multi-objective nor an asynchronous version of cost-aware, partial-evaluation optimization of linked models has been developed, and none has been demonstrated on a physical alloy workflow. Published tests of the linked-model methods use test functions and simulators [57, 58]. Multi-stage Bayesian optimization for self-driving laboratories was tested on synthetic and retrospective data with one objective [70]. Cost-aware batch optimization of refractory alloys treats measurements of unequal cost as separate sources rather than linked steps, in a computational campaign [71]. Closed-loop campaigns on printed alloys use one model and a fixed set of measurements [72]. Active learning has chosen compositions of multi-principal element alloys deposited by DED, scored on hardness [73], and a multi-fidelity model has combined an analytical heat-flow model with finite-element simulations to map DED settings [74]. Neither linked the steps or their costs. Alloys designed for printing by computation, such as an oxide-dispersion-strengthened NiCoCr alloy [75] and high-strength aluminum alloys [76, 77], were designed and then printed as single compositions, and a nickel superalloy has been designed with a neural network for laser repair [78]. Machine learning has refined powder-bed settings for 70-30 copper-nickel at the Naval Postgraduate School [79], and Bayesian optimization has tuned a gas atomizer [80]. Both tune the process for one fixed alloy. Standard multi-objective Bayesian optimization handles competing properties and batches [81] but treats the whole chain as one costly experiment. Printed copper alloys for seawater have used the cast or wrought compositions [8, 9, 10], and their corrosion studies characterize one alloy and one process at a time [28, 29, 37, 39]. A 2026 study adjusted the composition of NAB for laser additive manufacturing to improve its as-deposited strength and corrosion resistance [11], and adding 0.1% titanium to wire-arc NAB lowered its corrosion current by 16% [82]. Both show the payoff of designing for the process. Autonomous and high-throughput corrosion work has used a scanning droplet cell to screen coatings and has run fast screens before slower, more thorough tests [83, 84], but without deciding, candidate by candidate, when a long test is worth its cost. This project combines the linked network, the cost per step, physics computed for the process, several objectives including corrosion, and results that arrive out of order, and tests the combination on real deposits.

1.5 Preliminary results

Since the lab started at BYU in December 2025, we have set up a CALPHAD workflow on BYU high-performance computing, installed the ultrasonic atomizer, and completed a powder-to-part demonstration on a baseline aluminum alloy. Atomized powder, laser powder bed fusion builds, scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDS) results, and tensile data are in hand. The atomizer's induction module also melts copper alloys, up to its working limit of 1,300 °C. NAB and 90-10 copper-nickel melt about 250 and 150 °C below that limit, and 70-30 copper-nickel within about 60 °C of it [TODO Sterling: add the first copper powder (SEM image and size distribution) from the runs planned for the week of 19 October; attachments can be replaced until SPO submits on 28 October, and 3 pages of the 20-page limit are still free.] The DED facility of Task 2 has already melted copper-nickel powder spread in grooves, the pre-placed tracks that Task 2 uses as its first deposition step [TODO Sterling: ask Sam whether these tracks, and a photo, may be shown].

We also simulated the cost-aware method on a synthetic version of the workflow (Figs. 2 and 3), made for the aluminum work. With the crack screen ignoring zirconium, screening and then tuning

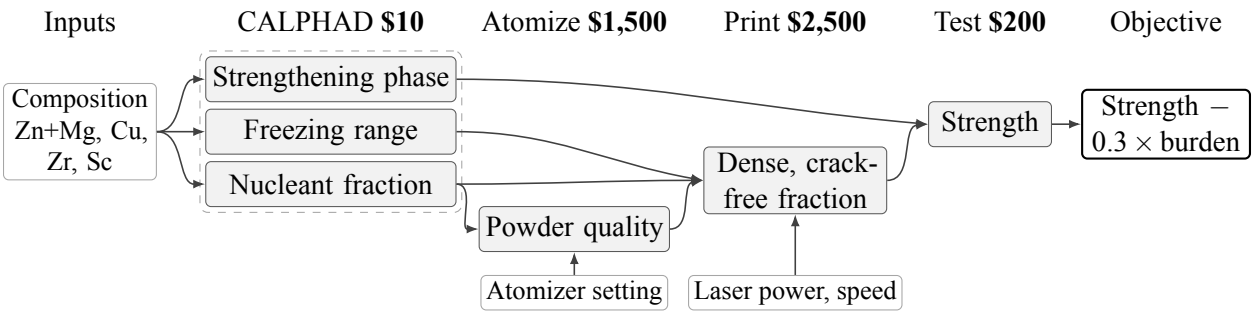


Figure 2. Synthetic workflow used for Fig. 3, built for the aluminum version of this workflow: a hand-written function for each box and a rough total cost (time, materials, supplies) for each step. Burden is a penalty on scarce elements such as scandium. The nucleant (Zr, Sc) cuts cracking, an effect the CALPHAD crack screen does not see. Task 1 replaces these functions with copper and DED ones.

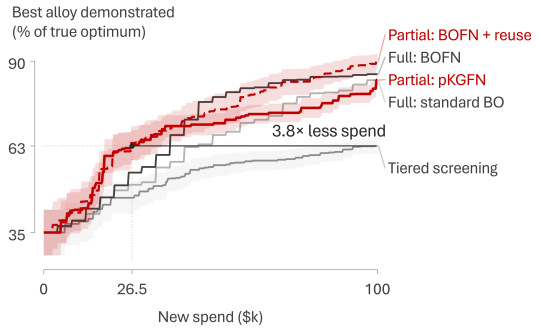


Figure 3. Best result found versus spend in simulations where the CALPHAD crack screen ignores zirconium (means of 20 campaigns, ±1 standard error). Red searches choose the next step; gray ones follow a fixed order. pKGFN is the cost-aware method [58] and BOFN the linked-model method [57]; reuse reprints earlier powders.

the process stalled at 63% of the optimum after $100,000. The cost-aware search, and a simpler one that reused powders, reached that level after about $26,500 (3.8 times less spend; 90% confidence interval 2.0 to 5.7). Searches that ran every candidate through the full chain caught up later, and screening kept pace when its physics was right. The simulation is hand-built and noise-free, with one objective, no corrosion branch, no process physics, and one experiment at a time, so it does not test the advances proposed here. It does show that letting the search choose the step can pay off when the physics model is incomplete. The DED workflow has the same structure: a physics screen that can miss effects, a powder step, a deposition step, and tests.

1.6 Method advances

Advance 1: several competing objectives. The published cost-aware method optimizes a single objective. A copper alloy for DED repair must balance deposit quality (dense, crack-free, and sound at the fusion line with the part it repairs), strength, ductility, and corrosion resistance in seawater, with no weighting known in advance. We will extend the cost-aware rule to several objectives. The expected improvement in one objective will be replaced by the expected growth of the set of best trade-offs (the Pareto front), measured by hypervolume as in multi-objective Bayesian optimization [81], and divided by cost as before. The difficulty is that each cheap measurement informs some objectives strongly and others weakly. Single tracks show cracking and dilution well but say little about strength or corrosion, and a polarization measurement says much about corrosion rate and nothing about strength. The rule has to account for that through the network. We will derive the rule, check it against brute-force estimates on small problems, and measure its

computing time on BYU high-performance computing.

Advance 2: results that arrive out of order. The published method chooses one experiment at a time and waits for its result. In this workflow a DED session deposits many specimens, microscopy results arrive days later, immersion tests run for weeks, and thermodynamic calculations finish in minutes. We will extend the rule to batches whose results arrive at different times, so that new experiments are chosen while others are still pending. Pending experiments will enter the models as simulated results drawn from the current predictions, a standard way to avoid choosing the same experiment twice [85]. We will also test how stable the choices are when a late result contradicts early ones. For example, a block that looked dense on its surface may show cracks under the microscope, or an alloy that passed the polarization screen may show selective attack after a month of immersion.

Advance 3: physics computed for the process. Standard CALPHAD inputs describe equilibrium, or solidification without diffusion in the solid, and ignore the thermal history of the process [86]. We will add inputs computed along the DED thermal history (Section 1.3): cooling rate through solidification, the peak temperatures of reheating, and the phases that CALPHAD allows at those temperatures. Hypothesis H3 tests whether these inputs make the models' predictions more reliable than equilibrium inputs, and whether equilibrium inputs beat data alone. The answer measures what incomplete, process-aware physics is worth inside a search, which sets how much simulation a campaign should buy before it buys experiments.

Advance 4: a physical workflow from powder to repair. We will demonstrate the advances on a physical workflow (Fig. 1) in which the search chooses the composition, the atomizer settings, and the deposition settings, and decides at each step whether a candidate goes on. After microscopy, specimens go to tensile tests, or to the corrosion screen and, for selected candidates, to immersion tests. Candidates that deposit well get a repair coupon: a deposit on cast NAB or wrought copper-nickel plate, checked for cracks, dilution, and hardness across the fusion line. Pre-placed tracks on the same plates come first. The demonstration includes real deposit-to-deposit scatter, outputs computed from microscopy images, and three inputs that enter at different steps.

Advance 5: checking what the screens discard. A cheap screen saves money only if it does not discard good candidates, and Section 1.2 shows that a short corrosion screen can also pass bad ones. Each round, the search will send a small random share of the candidates it would stop, about one in ten, on to the next step anyway. These audited candidates measure how often each screen is wrong, and they keep the models from learning only from candidates that the screens favored.

1.7 Hypotheses and rejection criteria

H1 (cost). The cost-aware, multi-objective search that handles pending results will reach a target set of trade-offs at lower total cost than standard batch multi-objective Bayesian optimization, which carries every candidate through the full chain. H1 will be rejected if, across matched campaigns with the same budget, the median cost to reach the target is not lower than the full-chain baseline's.

H2 (calibration). The predicted probabilities that a candidate will deposit dense and crack-free, fuse to the repair substrate without cracks, and stay close to its intended composition will match the observed rates on held-out deposits. H2 will be rejected if observed success rates differ from predicted probabilities by more than 10 percentage points.

H3 (process-aware physics). CALPHAD outputs evaluated along the DED thermal history will improve that calibration over CALPHAD outputs at equilibrium, which will in turn improve it over models trained on data alone. H3 will be rejected if the process-aware inputs do not reduce calibration error on held-out deposits compared with the equilibrium inputs.

H4 (pending results). Choosing new experiments while others are pending will reach the target in less calendar time than waiting for each batch to finish, at equal spend. H4 will be rejected if, in median calendar time, the version that works around pending results is no faster than the version that waits.

H5 (corrosion screen). Corrosion rates, breakdown potentials, and impedance from the one-hour screen, together with composition and microstructure, will predict which candidates pass a 30-day immersion test, defined as a mass loss no greater than that of the 70-30 copper-nickel reference and selective attack no deeper than [TODO Sterling: depth. Immersions of 21 and 30 days in salt solution found at most 1.8 to 3.4 μm in laser-deposited NAB [32] and more than 3 μm in cast NAB [87]. Options: a few micrometers, or no deeper than in the cast NAB reference]. H5 will be rejected if, among tested candidates (audited ones included), those that the model ranks in the top half pass no more often than those in the bottom half.

The targets, baselines, budgets, and number of matched campaigns will be fixed and posted before each test starts [TODO Sterling: confirm a public pre-registration site, e.g., OSF]. H1 and H4 are tested first on simulated and retrospective benchmarks, where many matched campaigns are affordable, and then in live campaigns, where only a few are.

1.8 Research plan

Task 1: methods (Ph.D. student 1). Derive and implement the multi-objective rule (Advance 1), the rule for pending results (Advance 2), the process-aware physics inputs (Advance 3), and the audit of screened-out candidates (Advance 5). Build simulated benchmarks that replace the functions of Fig. 2 with copper and DED ones, with noise, several objectives, a corrosion branch, and delays drawn from measured turnaround times. Build retrospective benchmarks from the lab's aluminum data and from the copper data that Task 2 adds each year.

Task 2: experimental platform (Ph.D. student 2). Powder for each candidate will be made in our laboratory. Oxygen-free copper bar is machined into cups that hold the charge. Nickel, iron, manganese, aluminum, and other additions are weighed in to 0.1 mg as pure metals or copper master alloys, about 0.5 kg per run, as in the published CuSn8 runs [88] [TODO Sterling: confirm the charge size after the first copper runs]. The charge is melted in a graphite crucible in the induction module of the ultrasonic atomizer (AMAZEMET rePOWDER) and poured through a 0.5 mm nozzle onto a vibrating plate, where it breaks into droplets. Nickel and iron react with graphite, so before melts that contain them the crucible is sprayed with a thin boron nitride coating, which lasts a few runs, and the melt is poured through a boron nitride nozzle, as the manufacturer advises. NAB has already been atomized from a graphite crucible on a rePOWDER [16]. Many alloys wet bare carbon-carbon composite poorly [88], so the plate is carbon-carbon with a molybdenum- or tungsten-based coating where the melt lands. The first part of each pour heats the plate until the melt wets it, and atomization is then continuous. With such plates on our model of atomizer, the manufacturer turned 80 to 90% of each charge of a tin bronze (CuSn8) into powder [88] [TODO

Sterling: confirm with AMAZEMET the plate and nozzle for NAB and copper-nickel; the published NAB runs used a bare carbon-fiber plate that left fibers in the powder [17]]. The median droplet size follows the wavelength of capillary waves on the plate, so it falls as the frequency to the power $-2/3$ [89]. The plates on our unit vibrate at 40 kHz, where Lang's relation gives a median near 47 μm for a copper melt. Measured powders are coarser. NAB atomized at 40 kHz had a median of 98 μm [17], about twice the relation's estimate, and much of the CuSn8 powder made at 40 kHz on our model fell between 63 and 150 μm [88]. Both suit the DED system described below, which takes powder of about 50 to 150 μm, sometimes up to 200 μm, and is usually fed cuts of 45 to 105 or 100 to 200 μm. Published blown-powder work on bronzes used 45 to 106 and 45 to 150 μm cuts [27, 90]. Each lot is sieved into those two cuts and checked for size distribution (laser diffraction), flow and apparent density with a Hall funnel [91, 92], shape, satellites, and stray plate fibers (SEM), and oxygen and carbon content [TODO Sterling: name the service laboratory: inert-gas fusion for oxygen, combustion for carbon]. The atomizer settings the search chooses are the melt temperature, the pour pressures, and the plate amplitude, and the share of each lot in each cut is an output of the powder step. With the gas flows and the laser spot, the size distribution sets how much of the powder the melt pool catches [14]. The search chooses the atomizer settings along with the composition, so the powder, not only the alloy, is designed for blown powder.

DED sessions will be bought from [TODO Sterling: facility name and location], which runs a FormAlloy X2 blown-powder laser DED system, at its external rates for machine time and labor. A typical session, four 25 mm cubes, takes about 3 hours of machine time and 2 hours of labor. [TODO Sterling: Confirm the laser (wavelength, power, spot size), its shielding (chamber or local argon), substrate preheat, turnaround, that powder left in the hopper comes back to us, and whether a smaller hopper can run a few hundred grams.] The facility plans on the melt pool catching about half the powder fed, and the hoppers of the X2 need about 2 kg of each powder. Since 2 kg takes about six atomizer runs, each composition first gets a cheaper deposition step that the facility has already used for copper-nickel. Powder from a first lot is spread in grooves machined in plates, including cast NAB and wrought copper-nickel plate, and melted by the same laser at several powers and speeds. These pre-placed tracks use a few grams of powder each and show cracking, porosity, dilution, and fusion to the repair substrates. They differ from blown-powder tracks in heat input and in how the powder reaches the melt pool, so the models treat them as a cheaper, less faithful measurement. The heat flow model is calibrated separately for each kind of track, and the audit of Advance 5 measures how often the tracks mislead. A composition whose tracks are sound can be scaled up to 2 kg. Its first blown-powder session deposits single tracks at several laser powers, travel speeds, and powder feed rates (melt-pool size, dilution, cracking, and the share of the powder caught in the melt pool), thin walls (cracking and heat buildup), and small blocks (density and porosity). Powders whose blocks reach 99% relative density can get a second session with blocks for sub-size tensile bars (ASTM E8/E8M) and corrosion coupons at the chosen settings, and a repair coupon (Advance 4). Commercial 70-30 copper-nickel and NAB powders, deposited the same way, are the baselines [TODO Sterling: confirm commercial powder sources, at least 2 kg of each; the facility can also supply powder at cost]. The search decides which compositions are scaled up at all, since a full lot takes about six atomizer runs.

Selected deposits will be examined by SEM, by EDS and inductively coupled plasma mass spectrometry for composition (manganese and aluminum can be lost to evaporation and oxidation [86]), and by electron backscatter diffraction (EBSD) for grain size, texture, and phase fractions, includ-

ing martensitic β' in the bronzes, at BYU. Replicate tracks and blocks at fixed settings will measure deposit-to-deposit scatter. Image analysis scripts will turn micrographs into porosity, crack length, dilution, grain size, and phase fractions with stated uncertainty.

Corrosion tests follow the same staging. The screen runs on the laboratory's liquid-handling gantry (CubXL, Ursa Laboratories), which is already driven by scripts, so the search can schedule it like any other step. The gantry fills small O-ring wells, 3 to 4 mm across, clamped on polished faces, with substitute ocean water [47], and a potentiostat dedicated to the loop measures open-circuit potential, linear polarization, impedance, and cyclic polarization [48, 93]. Small cells can add crevice attack at the seal and shift breakdown potentials [94]. Each spot is therefore photographed before and after, and the wells are checked against a conventional 1 cm^2 flat cell on the references [TODO Sterling: choose the potentiostat: a module unit with impedance on the gantry, cross-checked on a borrowed laboratory instrument, or the benchtop unit and flat cell in Field 11; see cubxl-electrochemistry.md]. Several wells fit on one specimen, so a repair coupon is measured in the deposit, at the fusion line, and in the substrate's heat-affected zone, the zone where welded NAB was attacked selectively [7]. The gantry then pipettes the used electrolyte of each well into a tube for inductively coupled plasma mass spectrometry, which shows which elements left the surface, for example aluminum from NAB or nickel from copper-nickel. ASTM G61 is written for iron-, nickel-, and cobalt-based alloys, so the project will report how its procedure behaves on deposited copper alloys. Wrought 70-30 copper-nickel and heat-treated cast NAB are measured as references in every round. Candidates that the search selects get 30-day immersion tests in capped vials on the gantry's deck. The gantry draws a few microliters of each solution at about 1, 3, 10, and 30 days for the same analysis, so the metal released over time enters the models as an intermediate result between the one-hour screen and the final mass loss. The coupons are weighed before and after [50, 51] and sectioned to measure the depth of selective attack; the leading candidates get crevice and longer immersion tests [TODO Sterling: confirm the durations, and natural or substitute seawater]. Tests in flowing seawater, which take months, are left to Navy laboratories for the final candidates (Section 1.10).

The search space was chosen the way the laboratory chose its aluminum palette: a literature review of each candidate element, then a check of its coverage in the CALPHAD database. Fifty elements from the periodic table were reviewed one at a time, including every element that appears in a commercial copper alloy, a copper alloy or welding filler specification, a deoxidizing practice, or the atomizer's crucible and plate materials. Each review asked whether the element helps seawater resistance, deposit soundness, or repair; what sets its upper limit; whether it can be melted and atomized below the atomizer's 1,300 °C limit without attacking the crucible or plate; and whether Thermo-Calc's copper database (TCCU7 [95]) assesses its systems. Eight elements are searched from the start (Table 1), with copper as the balance at no less than 65% and a calculated liquidus no higher than 1,250 °C. Zirconium and boron, tin and germanium in NAB, and one rare earth, lanthanum or cerium, which improved a 90-10 alloy in salt solution [96], are tested in Year 2. Rules tied to the aluminum content keep each candidate near known metallurgy. Above 7% aluminum, nickel must at least equal iron, as it does in the nominal composition of Navy NAB castings [2]. Below 1.5%, iron stays at or below 2.3%, the upper limit of the iron-rich 70-30 grade C71640, so that it remains in solid solution [97]. Chromium and niobium are used only where aluminum is at most 3%, and silicon above 0.1% only with niobium, as in cast 70-30 copper-nickel [98]. Retained β is not limited by a fixed formula; the process-aware CALPHAD inputs predict it (Advance 3).

The space includes the region between the two families, copper-nickel-aluminum alloys such as the high-strength MARINEL [99], where low-nickel candidates may lie. Beryllium and cadmium (toxic powder and fume), zinc (it evaporates in the melt pool and dezincifies in seawater), and the refractory metals (immiscible, and far above the melting limit) are excluded. Lead, sulfur, phosphorus, carbon, bismuth, and the other impurities are held at or below the welding-grade limits for copper-nickel [100]. All the searched elements but iron are on the 2025 U.S. list of critical minerals [55]; nickel dominates by mass and is carried as an objective.

Table 1. Elements searched from the start (wt%), with copper as the balance (at least 65%) and a calculated liquidus of at most 1,250 °C. "Cu-Ni only": used where aluminum is at most 3% (silicon: below 7%).

Element	Range	Role	Upper limit set by
Ni	0–30	Protective film in Cu-Ni; κ phases in NAB	Liquidus near the atomizer's limit; supply (an objective)
Al	0–9.5	Aluminum bronze; Cu-Ni-Al alloys	Retained β' martensite as deposited
Fe	0–4.5	Film in Cu-Ni; κ phases in NAB	Iron out of solution in Cu-Ni; coarse κ in NAB
Mn	0–4	Deoxidizer; partial nickel substitute	Evaporation and oxidation in melting and DED
Cr	0–0.7, Cu-Ni only	Resistance to fast flow [101]	Cr-rich particles; crevice attack in still water
Si	0–0.5, Cu-Ni only	Deoxidizer; strength with Nb	Silicides; weld cracking
Nb	0–0.8, Cu-Ni only	Weldability of cast Cu-Ni	Undissolved Nb-rich particles
Ti	0–0.5	Deoxidizer in Cu-Ni welding filler [100]	Oxide and nitride inclusions; crucible attack

Task 3: live campaigns and hypothesis tests (both students). Run live campaigns in which the search chooses each experiment and step. Compare it on equal budgets with random selection and with standard batch multi-objective Bayesian optimization over the full chain. Test H2 and H3 on held-out deposits, H5 on the audited and selected candidates, and H1 and H4 in matched campaigns.

Task 4: release and transition. Release the software, data, and models each year (Section 5). Write a short guide for applying the method to other alloys, deposition processes, and test sequences, aimed at Navy laboratory and depot engineers. For each final candidate, publish a powder specification, the deposition settings, and the tensile, corrosion, and repair-coupon data, and share them with Navy materials engineers, who can decide which candidates merit flowing-seawater and service tests.

1.9 Expected results and new knowledge

If successful, the project will produce:

1. The first cost-aware method of this kind for competing properties and staggered results.

2. A measured answer to how much a staged, cost-aware search saves over full-chain testing on a real alloy workflow.
3. A measured value of process-aware physics: how much CALPHAD evaluated along the DED thermal history improves predictions over equilibrium calculations and over data alone.
4. Corrosion data for copper alloys deposited by DED, and a measured link between a one-hour electrochemical screen and immersion tests that take weeks.
5. New copper alloy compositions designed for blown-powder DED, each with its powder specification, deposition settings, tensile properties, corrosion behavior, and repair coupons, using less nickel where the trade-offs allow.

Results that reject a hypothesis are also new knowledge. For example, if H1 fails, the data will show which step's cost or noise erased the advantage; if H3 fails, they will show which process effects the calculations missed; if H5 fails, they will show which microstructural features the screen missed.

1.10 Future Naval Relevance

ONR S&T Focus Areas. The primary focus area is Manufacturing, research area Integrated Manufacturing and Sustainment Technologies. The secondary focus area is Materials and Electronics, research area Sustainable and Operationally Robust Materials. Within Code 332 the project fits the program Materials & Processes for Additive Manufacturing [102]. That program lists five thrusts: "(i) rapid qualification frameworks; (ii) tailored AM materials and processes; (iii) large scale AM; (iv) additive repair and (v) AM and digital logistics" [102]. Its research challenges include to "develop and validate computational approaches for AM materials and processes for engineering applications" and to "develop large-scale AM processes for naval materials of interest" [102]. The project serves thrusts (ii) to (iv) with an alloy designed for a large-scale process that also repairs, and thrust (i) with a search method that decides which test to run next and states its confidence in the outcome. It is ICME for additive manufacturing, the approach ONR's Structural Metals program calls "an efficient rigorous framework for optimizing processing and manufacturing schemes" [103]: computation first, and experiments where they change a decision. It also fits two other Code 332 programs: Corrosion Science and Corrosion Control Technologies, which asks for a "fundamental understanding of corrosion mechanism in additive manufactured (AM) alloys" [104], and Computer-Aided Materials Design, whose thrusts include high-throughput screening concepts and informatics [105].

Copper alloys, castings, and repair. The slow production of large castings "has historically gated progress and throughput for both the Columbia and Virginia submarine programs" [106], and the Navy is replacing castings with printed parts. In August 2026 it set out "a streamlined, repeatable technical pathway to substitute legacy cast and wrought parts with 3D-printed metallic materials" [107]. A printed copper-nickel deck drain, made "to provide an alternative to traditional copper-nickel castings," has been installed on a new-construction submarine [108, 109], and the Navy's Maritime Industrial Base program has made copper-nickel ship-service ball valves by wire-arc deposition [110]. These parts use the cast compositions, as does ONR-funded research on wire-arc 70-30 copper-nickel [37]. A copper alloy designed for blown-powder DED, one that resists seawater as deposited, would let a shipyard or depot build a part that now waits on a casting,

or restore a worn part in place without the heat treatment the cast alloys need. ONR's corrosion program asks for this kind of work: "modify the chemistry of feed stock materials to enhance AM processing and corrosion resistance of additive-manufactured alloys" [104].

The longer horizon. The project designs for single-stream, pre-alloyed powder, which existing DED systems can use now. The same models can later choose blends of elemental and master-alloy powders fed from several hoppers and alloyed in the melt pool, which would let a depot make a new composition without a new powder lot. The guide of Task 4 will describe that extension. The method needs only a network of steps, their costs, and their measurements, so it also applies to wire-arc and laser-wire deposition and to other Navy alloys, including titanium. A plasma melting module for our atomizer (not part of this request) would extend the powder work to alloys that melt above the induction module's limit.

Navy engagement and people. The PI presented autonomous experimentation for alloy discovery at the Naval Surface Warfare Center (NSWC) Carderock Division on 14 October 2026 [TODO Sterling: name any Carderock contacts who agree to be listed]. The PI will apply to the Summer Faculty Research Program to spend a summer at a Navy laboratory [TODO Sterling: confirm; the 2027 application closes 14 December 2026]. The project will train two Ph.D. students and undergraduates in powder making, deposition, characterization, corrosion testing, and autonomous experimentation. We will encourage them to apply to the Naval Research Enterprise Internship Program and the SMART and NDSEG programs, which the NOFO names as goals of the YIP.

1.11 Project Schedule and Milestones

Table 2 lists the schedule by year. The award start date is 1 April 2027.

1.12 Goals and Benchmarks

Table 3 lists each goal and the benchmark used to measure progress. Progress will be reported at each ONR program review.

1.13 Gold Standard Science

Executive Order 14303 lists nine tenets of Gold Standard Science [111]. Such science is reproducible, transparent, communicative of error and uncertainty, collaborative and interdisciplinary, and skeptical of its findings and assumptions. It is also structured for falsifiability of hypotheses, subject to unbiased peer review, accepting of negative results as positive outcomes, and without conflicts of interest. The plan follows each point. Each hypothesis above has a stated rejection criterion, and the targets and baselines are fixed before each test. Every prediction carries an uncertainty, and the calibration tests (H2, H3) check whether that uncertainty is honest. The audit of screened-out candidates (Advance 5) tests the screens' own assumptions. Replicate deposits and reference specimens measure scatter so that it is not mistaken for a real effect. Code, data, and models are released each year so others can rerun every analysis. Results that reject a hypothesis will be published like any other result. The PI's record supports this: the PI co-authored a review of self-driving laboratories and led open benchmark and software projects (see Section 3).

2. Management Approach

This is a single-PI project with no co-PIs or subawards. The PI will lead the method development, set the campaign plans, and supervise all students. Ph.D. student 1 will lead Task 1 (methods,

Table 2. Schedule and milestones (Y = project year, starting 1 April 2027).

Year	Work	Milestone (month)
Y1	Derive the multi-objective rule; build simulated and retrospective benchmarks; build the process-aware physics inputs; first copper powder lots in the DED size range and their pre-placed tracks; deposit the commercial baselines and replicate them to measure scatter; set up the corrosion screen on wrought and cast references; automate image analysis.	M1: first copper powder lot sieved, checked, and melted in pre-placed tracks (4). M2: baseline deposits and scatter measured (7). M3: corrosion screen repeatability measured (8). M4: multi-objective rule passes small-problem checks (9). M5: benchmarks frozen and posted (12).
Y2	Derive the rule for pending results; first live campaign with tensile, corrosion, and repair branches; first 30-day immersions; test H2 and H3 on held-out deposits.	M6: first live campaign done (18). M7: H2 and H3 tested (22). M8: methods paper and software release 1 (24).
Y3	Matched live campaigns against the full-chain baseline; test H1, H4, and H5; release software, data, and guide; share powder, deposition, corrosion, and repair data with Navy engineers.	M9: matched campaigns done (32). M10: H1, H4, and H5 tested (34). M11: final release and report (36).

physics inputs, benchmarks, software). Ph.D. student 2 will lead Task 2 (powder, deposits, microscopy, tensile and corrosion tests). An undergraduate researcher will help make and check powder, prepare specimens, and run the corrosion screen. Both Ph.D. students share Task 3, so the methods are tested on the data they will face.

DED sessions are bought at the external hourly rates of [TODO Sterling: facility], and any outside tests from a commercial laboratory; neither is a subaward. Staff at the facility run each session to the settings the search chooses [TODO Sterling: if Sam agrees to be named as an unfunded collaborator, say so here and list Sam under current and pending support (NOFO p. 38)]. About 14 blown-powder sessions and 6 sessions of pre-placed tracks a year fit the budget [TODO Sterling: confirm in budget.py; replacing the $18,000 potentiostat (Field 11) would add about 14 blown-powder sessions in Year 1]. All other work is at BYU: the PI's atomizer laboratory, the BYU Microscopy Center (SEM, EDS, EBSD), and BYU high-performance computing. The team meets weekly to review new results and choose the next experiments. A shared data pipeline records every step, so the search always sees the latest results. The schedule and milestones are in Table 2.

Risks and responses. If the multi-objective rule is too slow to compute, we will use a cheaper approximation and measure the loss on the simulated benchmark. If deposit scatter is large, more

Table 3. Goals and the benchmarks that measure them.

Goal	Benchmark
Multi-objective cost-aware rule	Matches brute-force estimates on small problems; finds a larger set of trade-offs per dollar than full-chain multi-objective search on the simulated benchmark.
Rule for pending results	Same final quality as the one-at-a-time rule in less calendar time on the simulated benchmark with measured delays (H4).
Calibrated predictions	Observed success rates within 10 percentage points of predicted probabilities on held-out deposits (H2).
Process-aware physics	Lower calibration error with thermal-history inputs than with equilibrium inputs, and with equilibrium inputs than without (H3).
Powder for blown-powder DED	Every deposited composition made in-house, sieved to the DED size range, with its size distribution, flow, and oxygen and carbon content recorded.
Corrosion screen	Top-ranked candidates pass the 30-day immersion test more often than bottom-ranked ones (H5).
Lower cost on real deposits	Lower median cost to reach the target than the full-chain baseline in matched live campaigns (H1).
New alloys	At least [TODO Sterling: number] new compositions deposited dense and crack-free, with tensile, corrosion, and repair-coupon data, beyond the commercial baselines.
Open release	Software, data, and models posted each year with tests and documentation.

replicates will be added and the scatter will enter the models directly. If a composition cannot be atomized, for example because it melts too close to the module's limit, the search records the failure as a result, since predicting which compositions can be made is part of the task. If a plate's coating wears and the melt stops wetting it, the plate is replaced; the tungsten-based coating, the best balanced in the manufacturer's comparison [88], is the alternative. If melts that contain nickel or iron still pick up carbon, which each lot's carbon analysis will show, the crucible will be recoated before every run instead of every few. An alumina sealing rod and a molybdenum-alloy plate [112] will also replace the graphite rod and the carbon-carbon plate. If a lot flows poorly, the search sees that at the powder step and can change the atomizer settings before paying for a deposit. The pre-placed tracks on cast NAB and copper-nickel plate show early whether the first layer fuses to thick plate; if it does not, the repair coupons will be preheated or made thinner. If making 2 kg lots slows the campaigns, the pre-placed tracks will carry more of the screening, and fewer compositions will be scaled up. If DED turnaround is slow, the rule for pending results is designed for exactly that case. If the electrochemical screen predicts the immersion tests poorly (H5 rejected), the search will rely more on immersion, at higher cost, and the audited candidates will show which features the screen missed.

3. Principal Investigator Qualifications

Sterling G. Baird received a Ph.D. in Materials Science and Engineering from the University of Utah in August 2023. He has been an Assistant Professor of Mechanical Engineering at BYU since December 2025, his first tenure-track position. Before BYU, he was Director of Training and Programs and then a Principal Investigator and Staff Scientist at the Acceleration Consortium, University of Toronto, where he built the Acceleration Consortium's development laboratory, including about $600,000 of equipment.

The PI's work combines Bayesian optimization with laboratory hardware. He developed Honegumi, which helps experimentalists set up Bayesian optimization, consulted for Meta on the version 1.0 release of the Ax platform, and organized an international Bayesian optimization hackathon. He co-authored the structural materials section of the *Chemical Reviews* survey of self-driving laboratories and led a review of low-cost self-driving laboratories. His graduate work used Bayesian inference with propagated uncertainty to model grain boundaries. He also designed a semi-automated induction furnace for temperatures up to 2500 °C. At BYU he supervises eight undergraduate researchers and has set up the CALPHAD workflow, the atomizer, and the powder-to-part demonstration described in Section 1.5. [TODO Sterling: add any corrosion or electrochemistry experience; if there is none, say how the students will learn the corrosion methods, e.g., a BYU colleague's training or a Navy laboratory visit. Add any DED or welding experience.] The CV lists all publications and presentations.

Principal Investigator's Curriculum Vitae

[TODO Sterling: insert the full CV PDF here.]

Letter of Support

[TODO Sterling: insert the signed letter from the BYU official here as a PDF page (draft in support-letter-draft.md; it names aluminum, so revise it for this version).]

4. Responsibility

Resources. The PI's laboratory at BYU holds the ultrasonic atomizer and the equipment for powder handling and specimen preparation [TODO Sterling: list the main items bought with start-up funds]. A potentiostat for the corrosion screen is requested in Year 1. The BYU Microscopy Center provides SEM, EDS, and EBSD, and BYU provides high-performance computing. DED sessions are bought as a service, as described under Task 2. Tensile tests use [TODO Sterling: name the BYU load frame or test service].

Ability to comply. The PI's current commitments are teaching, start-up research, and the pending requests listed under current and pending support. The effort proposed here fits within those commitments.

Performance history. [TODO Sterling: list any federal awards managed as PI or key person; if none, state that this would be the PI's first federal award as PI and that BYU Research Administration manages the award.]

Integrity and business ethics. The PI has no findings of research misconduct and follows BYU's policies on research integrity and conflicts of interest. [TODO Sterling: confirm.]

Qualifications and eligibility. The PI received his Ph.D. after 1 January 2019, holds his first full-time tenure-track position, and has not received an ONR YIP award. [TODO Sterling: confirm U.S. citizenship or permanent residency.] BYU is registered in SAM.gov (UEI JWSYC7RUMJD1, CAGE 52534) [TODO Sterling: BYU's SAM registration is listed as ending 4 March 2027, before the award start; confirm that RAO will renew it].

Organization and controls. BYU Research Administration manages federal awards, with accounting, purchasing, and property control systems [TODO Sterling: confirm wording with BYU Research Administration]. Laboratory safety follows BYU Risk Management and Safety procedures. Metal powders are handled under the lab's written powder safety plan; nickel-bearing powders are handled in a fume hood or glove bag with respiratory protection [TODO Sterling: confirm the plan and the handling steps for nickel powder]. Corrosion tests use salt solutions; the dilute acids used to clean immersion coupons [51] are handled in a fume hood under BYU chemical safety procedures.

5. Data Management Plan

Types of data, software, and other materials. Thermodynamic and heat flow calculation results; atomizer settings and powder size, flow, density, oxygen, and carbon records; settings and session logs for pre-placed tracks and DED deposits; optical and electron micrographs, EDS spectra, and EBSD maps; tensile and hardness data; polarization and impedance data, immersion mass losses, and cross-sections of corrosion attack; the search software, benchmark problems, and trained models.

How the data will be acquired. Calculations run on BYU high-performance computing. Powder is made in the PI's laboratory. DED sessions are bought as a service (Section 2). Microscopy, corrosion screens, immersion tests, and mechanical tests are done at BYU.

Time and location. Data are collected at BYU and at the DED facility from April 2027 to March 2030.

Processing. Scripts kept under version control turn raw images and test records into derived quantities (porosity, crack length, dilution, grain size, phase fractions, strength, corrosion rate, attack depth) with stated uncertainty. Each derived value links to its raw files.

File formats and naming. Open formats: CSV and JSON for tables, TIFF for images, and the instruments' native formats alongside open exports. Each sample receives an identifier that encodes powder lot, deposition session, and specimen, used in every file name.

Quality assurance and control. Replicate deposits measure scatter; reference specimens of wrought 70-30 copper-nickel and cast NAB are measured in every round of corrosion screening; scripts are tested; and the same benchmarks are rerun before each software release.

Existing data. The retrospective benchmarks use the lab's own aluminum data from 2026, with its origin recorded in the metadata.

Standards. Metadata will record composition, powder properties, process settings, instrument settings, and the ASTM method behind each test in a documented schema [TODO Sterling: choose a schema, e.g., one used by the NIST AM Bench or Materials Data Facility].

Preservation. Data and software will be deposited in public repositories (for example, Zenodo with a DOI, and GitHub for code) at each annual release and kept for at least 10 years.

Balance of value and cost. Raw session logs and micrographs will be kept in full; very large intermediate files that scripts can regenerate will not be archived.

Release restrictions. Results are expected to be fundamental research. Any data that cannot be cleared for public release under DoD Instruction 5230.09 will not be made public.

Current and Pending Support (summary)

Current. [TODO Sterling: list current awards, if any, including BYU internal funds that must be reported under the common form.]

Pending. AFOSR FY2027 Young Investigator Program, white paper submitted 9 October 2026 (NOFO FA955026S0003, Dynamic Data and Information Processing, Dr. Erik Blasch). Title: "Dynamic Data-Driven Selection of Experiments for Laser-Printed Aluminum Alloys." Up to $150,000 per year for 36 months. The two share the search method. This proposal applies it to copper alloys for blown-powder DED, with corrosion and repair branches; the AFOSR work applies it to aluminum alloys for laser powder bed fusion, so the experimental work does not overlap. [TODO Sterling: state how the method work (Task 1) would be divided if both are selected. Add any other proposals submitted before 30 October, and list collaborators, even unfunded ones (NOFO p. 38).]

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